Research Tracks and Research Plan

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1 Previous Research Experience

Since arriving at KU Leuven in Belgium in 2019, I have been involved in several EU projects closely collaborating with industry, with a total budget of approximately €16.55 million. Efforts have been made to translate project outcomes into practical applications, which are a core and crucial assessment criterion.

I'm happy to be involved and coordinate the work across different research groups that currently consists out of 10+ PhD students and 3 post-docs. My research activities and worked in the past on the can be divided in 3 tracks, manufacturing, in-process inspection, AI applications:

I'm more than happy to participate and coordinate the collaboration among various research groups for interdisciplinary studies, which comprise over 10 PhD students, 4 post-docs, and 6 PIs. My research endeavors can be categorized into three main tracks: *precision manufacturing*, *process monitoring*, and *AI application for industrial use*. Each of these tracks will be further elaborated in the following sections.

1.1 Precision Manufacturing

The research journey in machining began in 2013 when I embarked on my master's degree. Accumulating substantial experimental experience in specialty processing, I published "Experimental Study on Side Wall Insulated Electrode Pulse Electrochemical Machining" in *Modern Manufacturing Engineering* in 2015, along with a series of patents (details available in the resume). As I pursued my doctoral degree, my focus shifted from predominantly applied research based on process experiments to fundamental research on process regularities. In 2018, as the lead author, I published "Modeling and Simulation of the Material Removal Process in Electrolyte Jet Machining of Mass Transfer in Convection and Electric Migration" in *Procedia CIRP*. This paper established an ion kinetics model for the process of electrochemical jet machining, capable of describing a series of physicochemical reactions at the ion level during machining.

As my research delved deeper, in 2020, as the lead author, I published "Fabrication of surface microstructures by mask electrolyte jet machining" in the *International Journal of Machine Tools and Manufacture* with **an impact factor of 14**. This paper addressed the fabrication of large-area surface microstructures, developing a novel process of lithographic template jet electrolysis. Through a multi-physics field model, the mechanism of this process was thoroughly investigated, exploring the theoretical limits of consistency in processing surface microstructures on a large scale, and proposing potential methods for further enhancing the process. In 2020, further research on the lithographic template jet electrolysis process led to the publication of "Fast Fabrication of Complex Surficial Micro-Features Using Sequential Lithography and Jet Electrochemical Machining" in *Micromachines*, with me as the lead author. This paper not only analyzed machining results using multi-physics field models but also employed a probability density distribution model to analyze the statistical differences in machining results and infer the causes of these differences. This led me to gradually realize the limitations of purely physical modeling and the importance of process monitoring in subsequent doctoral and postdoctoral research.

Beginning in 2021 with postdoctoral research, I gradually shifted from physics-driven models to data-driven models in fundamental research, but never completely abandoned physical models. Instead, relying on my experience in data, I attempted to further improve the computational performance of multi-physics field models. In 2023, as the lead author, I published "Experimental and Numerical Investigations on Fabrication of Surface Microstructures Using Mask Electrolyte Jet Machining and Duckbill Nozzle" in the *Journal of Manufacturing Science and Engineering*. When conducting engineering research based on physical models, limitations in measurement methods and computational costs often necessitate varying degrees of simplification of the physical model in practical applications. This paper introduced a method of "sub-modeling first and then assembling parts through continuity equations," which, based on the importance of different parts, applies varying degrees of simplification to different parts, thereby significantly reducing the total computational cost.

1.2 Process Monitoring

In the latter stages of my doctoral research, process monitoring became one of my key research focuses. In 2020, I collaborated with Professor Dominiek Reynaerts' team at KU Leuven and co-authored a paper titled "Tool Wear and Material Removal Predictions in Micro-EDM Drilling: Advantages of Data-Driven Approaches," published in *Applied*

Science. This marked the first instance of predicting tool wear in electrical discharge machining (EDM) using machine learning models on a commercial EDM machine, the Sarix® SX-100-HPM, followed by deploying the model on the machine and achieving feedback control.

In 2021, I, as the lead author, presented a paper titled "Data-driven framework of ECM: A machine learning model for profile prediction" at the *Proceedings of the 17th International Symposium on Electrochemical Machining Technology (INSECT 2021)*. Leveraging various machine learning models and process signals in electrochemical machining, we predicted machining outcomes. As the research progressed, the prediction accuracy improved, leading to the publication of "Profile prediction in ECM using machine learning" in *Procedia CIRP* in 2022, with me as the lead author. Subsequently, by optimizing the computational costs of relevant algorithms, this achievement was deployed in cloud as **an web application** on KU Leuven's cloud servers in 2024.

Apart from applications in precision manufacturing, I co-authored a paper titled "Experimental Investigations into Machining Characteristics of Niobium Carbide Cermet with µECM/hybrid laser-µECM" published in *Procedia CIRP* in 2022. Utilizing data from the machining process, we analyzed the process characteristics of machining composite materials.

Given the current technological constraints preventing direct sensor deployment, I proposed the use of a "virtual sensing" approach. This method simulates physically unmeasurable quantities using physical models. In 2022, I co-authored a paper titled "A 'virtual sensing' framework for µECM/hybrid laser-µECM for monitoring interelectrode gap conditions" published in *Procedia CIRP*.

In 2023, as projects I was involved in transitioned from non-traditional to traditional machining, I engaged in academic exchanges and collaborations with the Aerospace Engine Manufacturing team at Northwestern Polytechnical University. Leveraging milling process signals, we predicted surface roughness of aerospace engine blades, obtaining a "panoramic information map" of blade surface roughness. These findings were published in *Mechanical Systems and Signal Processing* in 2023.

1.3 AI applications for industrial use

In 2022, I joined the DTAI team at KU Leuven's Department of Computer Science to further pursue interdisciplinary research in AI and manufacturing. In 2023, as the lead author, I presented a paper titled "An unsupervised learning method for pulse classification in EDM" at the *Proceedings of the 23rd International Conference of the European Society for Precision Engineering and Nanotechnology*. This paper transplanted algorithms from the field of speech recognition to the realm of electrical discharge machining (EDM) for waveform recognition. It introduced a dynamic time convolution method for calculating similarity between signals of varying lengths and amplitudes and subsequently clustered these signals using unsupervised learning. Due to its applicability to signals of varying lengths and amplitudes, and its efficiency and generality compared to traditional threshold methods and supervised machine learning approaches, this method was optimized for computational costs and parallel computing and finally released as **standalone software** in 2024 (copyright owned by KU Leuven).

In 2023, I presented a poster titled "Representation of machining state and optimization of rough milling efficiency using machine learning" as the lead author at the *2023 Flanders Make Scientific Conference*. This work digitized all states of the milling process and optimized machining strategies for rough milling using machine learning. This achievement garnered significant attention from European manufacturing companies. Building upon this research foundation, in 2024, along with Professor Mathias Verbeke and others, I co-drafted a proposal, **AutoCAM_SBO** with a total budget of **2.75 million Euros**. This proposal received the highest score in the penultimate round of review, with one reviewer describing it as "a no-brainer" with the highest score among all submissions.

2 Research Plan

2.1 Co-pilot for Function-Driven Product Design: Automated Generation of 3D Designs Tailored to Specific Functional Requirements

In the realm of product design, the pursuit of innovation and efficiency has long been intertwined with the advancement of technology. With the emergence of Generative Artificial Intelligence (AI), a transformative paradigm shift has occurred, revolutionizing the way designers conceive, iterate, and actualize their creations. This shift is particularly pronounced in the realm of 3D design, where Generative AI offers unprecedented capabilities for the automated generation of designs tailored to specific functional requirements. Traditionally, product design has been a labor-intensive process, often constrained by the limitations of human imagination and expertise. Designers are tasked with reconciling numerous considerations, from form and aesthetics to functionality and manufacturable. However, the advent of Generative AI has introduced a novel approach, empowering designers to harness the computational prowess of machine learning algorithms to explore vast design spaces and uncover innovative solutions.

Central to the promise of Generative AI in product design is its ability to generate 3D designs that are not merely aesthetically pleasing, but also aimed for specific functions. This aligns with traditional design methodologies, where functional considerations typically take precedence, yet creative exploration remains a vital aspect of the design. In this thesis work, the primary focus will be on the generation of part designs tailored to specific functions, this entails generating 3D designs based on functional requirements, encompassing elements such as "rotation," "translations,"

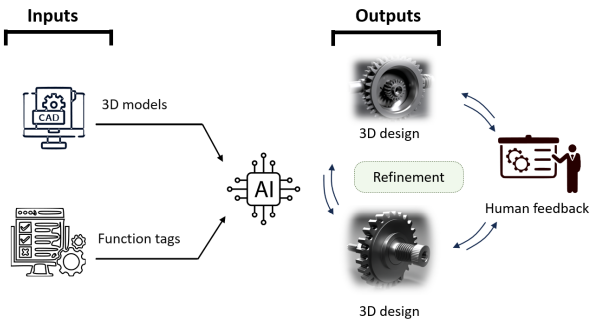

Figure 1: Overview of the generative AI-driven product design process

or an assembly like "gear train". This sets the groundwork for eventually creating fully production-ready designs. Then, the research will be guided by the following objectives: 1) Construct a training dataset from publicly available 3D assets sourced from CAD websites; 2) Annotate the training dataset with tags that encapsulate the respective functions of the 3D assets. 3) Adapt the current Generative AI algorithms to the product design workflow; 5) Evaluate the performance and effectiveness of the Generative AI model through rigorous testing and analysis; 6) Iteratively refine the design through human feedback

2.2 Context-Dependent Machine Learning for Monitoring Purposes in Additive Manufacturing

Monitoring additive manufacturing processes such as Laser Powder Bed Fusion (LPBF) is an important asset to early stop or adjust the configuration of the printing of an object when it did not occur according to the desired standard. Although different sensing modalities can be used for monitoring in this work acoustic signals are targeted. To search for signal patterns in these in-process signals that link to abnormal printing machine learning algorithms are used. However, as the process configuration (e.g. laser power and scanning speed) in LPBF, which impacts the acoustic signals, is frequently adjusted there is a need to make the machine learning aware of these changes in configuration.

One approach that could be taken in this thesis work is to divide the model into segments: one part agnostic to the processing configurations, while the other remains gnostic to them. Consequently, when alterations occur in the process configurations, only the segments linked to these configurations will undergo retraining. A strategy of 2 training stages will be studied to achieve this goal.

Stage 1 Employing an encoder to encode inputs, with the green layers (input encoder) remaining fixed after initialization. Solely the in-process data serves as inputs to train the machine learning model. The process settings will be initialized to default values (ones/zeros) as placeholders. This step ensures that the grey layers remain agnostic to the process configurations.

Stage 2 Utilizing both process configurations and in-process data as inputs. Here, the grey layers' weights remain frozen to maintain agnosticism towards process configurations. However, the green layers are trained to adapt to various process configurations.

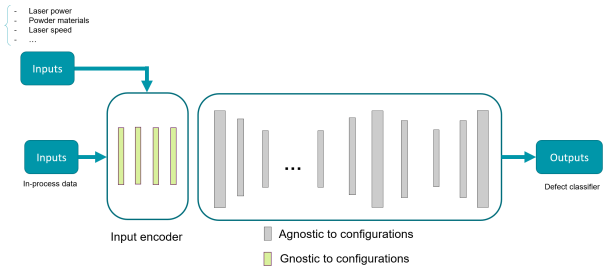

Figure 2: Schematic of Context Dependent Learning with Process Settings and In-process Data

The research will be guided by the following objectives: 1) Construct a training dataset from the available LPBF data; 2) Design a proper framework for machine learning models that can accommodate two-stage training; 3) Validate and fine-tune the model based on performance metrics and feedback from LPBF data; 4) Evaluate the effectiveness and efficiency of the 2-stage training approach in achieving the goal of adapting the model to varying process configurations.